

[DC] Remapped Interaction Detection by Naive Users in Virtual Reality

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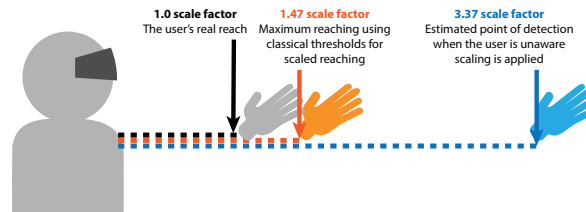


Figure 1: Summary of findings from Experiment 1 (completed work).

ABSTRACT

Remapped interactions have been often evaluated under the lens of detection; if we use these techniques at undetectable levels, we will not risk distracting the user, breaking immersion, or increasing cybersickness. However, research has relied heavily on methods that induce behavior and grant knowledge to participants that conflict with actual end-users. The goal of this dissertation is to evaluate the detection of these techniques by naive users, who do not know they are being used and are focused on other tasks to guide the usage of these techniques outside the laboratory.

Keywords: virtual reality, interaction techniques, human perception

1 INTRODUCTION AND RELATED WORK

Input remapping techniques in virtual reality (VR) provide many benefits to users, including but not limited to: increased use of natural movements to interact with their virtual environments such as reaching [5] and walking [6], easing fatigue [4], and facilitating interactions with real-world objects [1]. These techniques introduce proprioceptive discrepancies between a user's vestibular and visual systems. Thus, the detection of these techniques has been often considered; if we understand when this difference is undetectable to users we can limit their application to these ranges and reduce distraction or breaks in immersion while using remapping techniques.

Current methods for assessing detection largely rely on methods from classical psychophysics such as two-alternative forced-choice (2AFC) and Yes/No tests to determine detection thresholds. These methods have been applied to many varieties of remapped interaction techniques to produce detection thresholds that can be used to limit the technique to undetectable values (e.g., [2, 3, 7]).

However, the design of VR detection studies using these classical psychophysical methods often introduces elements to the experience that are not representative of actual VR application usage by end-users. Primarily, a subject in an experiment must necessarily gain prior awareness of the stimuli (e.g., the technique), which causes them to yield thresholds that are much more restrictive than can be attained when users are unaware of the technique's application. This level of technique awareness is not representative of naive end-users who, when using VR, may not be aware of the techniques they are using (and, ideally *are not aware* if we seek to minimize detection). Ultimately, it is uncertain just how limiting and restrictive thresholds derived from classical methods are.

If we were able to assess detection without users being aware a remapped interaction was applied, we could increase the range of

usable technique intensities and the utility of remapped interactions. This is the first goal of my dissertation: designing and evaluating a methodology to estimate detection thresholds of remapped interaction techniques when participants are unaware a remapped interaction is being used.

Also important is a qualitative analysis of the user experience when technique detection occurs. Detection thresholds, as estimated by 2AFC methodologies, are still detectable to users approximately 25% of the time (these are defined as the points on a fit curve where participants were correct in the classification 75% of the time). It is uncertain how naive users, who use these techniques without the guidance of a supervising experimenter, perceive and understand these observed differences in their physical and virtual movements induced by remapping. Do they become distracted? Do they identify the change as helpful or harmful? Can they understand and explain *how* their movements are being modified, or solely that a modification is occurring? Do they perceive it as intentional, or as a glitch in the system?

In this work, I will conduct studies with two families of techniques: remapping of the virtual hand via scaling the hand speed, and remapping of head rotation via amplifying lateral turning. These are chosen to represent two classes of techniques; the former technique solely manipulates a user's body movements, while the latter affects a user's view of the virtual environment. While many other remapping techniques exist, it is necessary to limit the breadth of techniques included in this dissertation for practical purposes.

2 PROPOSED DISSERTATION WORK

Previously, I have authored three papers using classical psychophysical methods [2, 3, 8]. These prior publications investigated two hand-remapping techniques for reach (scaled hand motion [3] and hand offsetting [2]) and one view-manipulation technique for natural walking (strafing gain [8]).

My work will continue in this direction, and consider the use of remapped interactions by naive users who either may not know the techniques are being used, not expecting their use, or do not even know such techniques exist. I will evaluate both exocentric manipulations (e.g., those modifying a user's body movements) and egocentric manipulations (e.g., those modifying the user's view). Users are more sensitive to the latter, regarding not only detection but also sickness, which may lessen any impacts that awareness may have. Thus, it is important to assess both.

2.1 Guiding Research Questions

This work is guided by two main questions:

- **RQ1:** How does the detection of a remapped interaction technique differ based on user awareness of the technique's application?

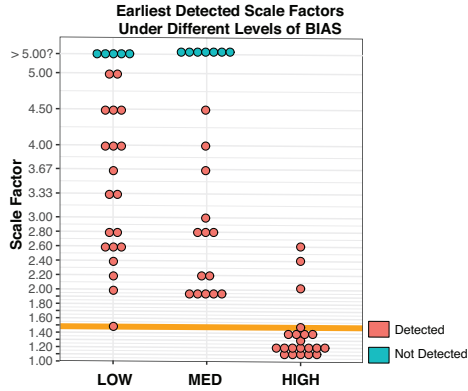


Figure 2: Detected scale values by levels of prior awareness (regarding technique) from Experiment 1. In the *HIGH* condition when participants knew the remapping occurred, detection happened at much smaller scales than conditions where they did not know.

- **RQ2:** How does the detection of remapped interactions affect the user experience of naive users?

2.2 Experiment 1: Effects of Technique Awareness on Remapped Hand Detection

This study has been completed and partially addresses **RQ1**. I evaluated the detection of scaled hand motion by users who had no prior awareness that the technique was applied. This same technique was previously investigated in [3] using 2AFC, though I also conducted a separate 2AFC analysis due to the different task and scale ordering in this experiment.

To do this, I developed a deceptive study methodology to elicit when naive users detect a hand-remapping technique. This was a between-subjects study with participants in three groups with different levels of prior awareness: *LOW* (fully unaware users without any knowledge their hand is remapped), *MED* (partially aware participants that are asked to look for something that feels unnatural), *HIGH* (participants that know their hand speed is changing and are asked every trial to make a judgment).

With $n = 77$ participants, I identified several interesting results. In our Yes/No condition (when participants knew their hand speed would sometimes be faster) we identified a threshold at 1.47 times the user's normal hand speed which is similar to that of [3]. However, in cases where the participants had no prior knowledge of their hand speed changes detection happened on average at a significantly higher speed of 3.37 times their normal hand speed. Additionally, approximately 1-in-5 participants who did not know any remapping occurred never noticed their hand speed increasing at our max speed of 5.0 times the normal speed.

2.3 Experiment 2: Effects of Technique Awareness on Remapped Head Detection

This planned future study will expand on and refine the methodology used in Experiment 1 to assess the detection of a different technique (head rotation gain) and also addresses **RQ1**. It is important to examine an egocentric view-manipulation technique because users are more sensitive to these methods than hand-manipulation techniques (regarding both detection and sickness). These methods can be used to support interactions such as natural walking to navigate an environment. This experiment will not only show the extensibility of the method developed in Experiment 1 but will also indicate if the lack of awareness of these techniques exhibited by naive users can produce similar decreases in detection as was identified for an exocentric manipulation used in Experiment 1.

2.4 Experiment 3: Naive User Understanding of Detected Remapping

This dissertation will culminate with an investigation of the user experience when a technique is detected to address **RQ2**. While we may aim to limit the use of these techniques to undetectable ranges, we can not guarantee that a user will never notice the technique. In these cases where, despite our best efforts, naive users do detect remapping has occurred it is uncertain what they will think of it.

Similar to Experiments 1 and 2, participants will unknowingly complete tasks with different intensities of remapping being applied. Each participant will experience 1) no remapping, 2) classical threshold remapping, 3) remapping at values from Experiments 1 and 2, and 4) extreme remapping. Both scaled hand movement and amplified turning will be evaluated as a between-subjects factor.

After each experience, I plan to utilize a series of questionnaires to assess outcomes such as embodiment, presence, distraction, cognitive load, etc. A semi-structured interview will also be used to determine if the remapping was identified and what the participant thought of it. Afterward, I will code and classify responses to the interview questions to identify trends in user understanding and thoughts toward remapping. This will inform our understanding of what happens to users when they detect unexpected remapping.

3 CONCLUSION

At the IEEE VR 2024 Doctoral Consortium, I hope to discuss the following topics:

- What utility is there in targeting VR researchers/developers as well as naive users in Experiment 3?
- Which outcomes, as they relate to impacts on the user when detection occurs, are most interesting to investigate in Experiment 3?

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